BCG X · MOCK INTERVIEW

Pack A — MediCare Plus

Hospital Readmission Prediction

DOMAIN	PROBLEM TYPE	DURATION
Healthcare / Risk	Binary Classification	55 minutes total

⚠️ **INTERVIEWER ONLY —** This guide and the coding solutions are for your eyes only. José should not see this document before the mock.

Interview Flow

Block	Your Role	Time	Cumulative
Welcome	Ask: *"Tell me about yourself — what brings you here?"*	2 min	2 min
Live Coding	Share problems. Watch screen. Ask comprehension question verbally.	15 min	17 min
Case: Setup	Read the scenario aloud. Give him time to take notes.	3 min	20 min
Case: His Turn	He drives. You reveal info when asked. Challenge his choices.	25 min	45 min
Synthesis	*"Can you summarize your recommendation?"*	3 min	48 min
Q&A	He asks you questions. Improvise as a DS at a top firm.	5 min	53 min
Debrief	Score the rubric together. One strength, one area to improve.	10 min	63 min

Case Scenario

Opening Script *(Read Aloud — 2 minutes)*

> *"Our client is **MediCare Plus**, a network of 15 hospitals across central Mexico with about 120,000 admissions per year. They're facing a problem with patient readmissions — patients who are discharged but return to the hospital within 30 days.*
>
> *Their current 30-day readmission rate is **18%**, well above the national benchmark of 12%. Each avoidable readmission costs the hospital an estimated **$8,500** in unreimbursed care, and regulators are starting to tie accreditation scores to readmission performance.*
>
> *MediCare Plus has asked a consulting team to help them **predict which patients are at risk of readmission** so they can intervene before discharge. They want a solution they can deploy across all 15 hospitals within 6 months."*

Information to Reveal — Only When Asked

Rule: Don't volunteer information. If he doesn't ask, don't tell. The whole point is testing whether he asks the right questions.

If He Asks About…	What to Say
Discharge process	Doctors use clinical judgment — no standardized risk scoring. Some departments do informal follow-up calls, but it's inconsistent.
Data / what's available	**Show Exhibit 2.** EMR system exists but completeness varies. Demographics and clinical history: solid. Social determinants (income, transportation, caregiver): only 41% of patients. Clinical notes: 60% unstructured.
Readmission patterns	**Show Exhibit 1.** Oncology and Cardiology are worst (>20%). Orthopedics and General Surgery are closer to benchmark. Nobody's investigated why.
Class balance	18% readmission — roughly 82/18 split. Imbalanced, but less extreme than fraud.
Interpretability	The CMO insists clinicians need to understand **why** a patient is flagged. They won't act on a black box.

Intervention budget	They can afford enhanced discharge planning for 15% of patients ($500/patient). Targeting matters.
Success metrics	Target: readmission rate below 14% in year one. That's roughly $5M saved in unreimbursed costs.

Exhibits *(Screen-share these at the right moment)*

When to show: Only when José asks about readmission patterns (Exhibit 1) or data availability (Exhibit 2). See the table above.

EXHIBIT 1 — Reveal when asked about readmission patterns

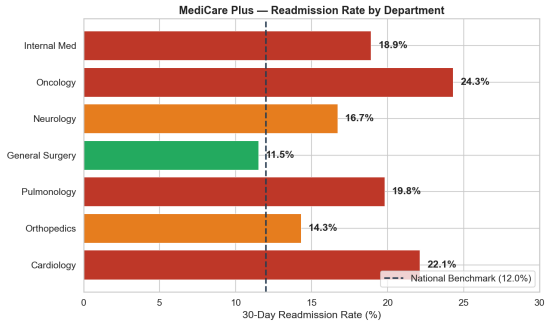

EXHIBIT 2 — Reveal when asked about data / features

MediCare Plus — Available Data Features

Feature Category	Examples	Data Completeness
Patient Demographics	Age, gender, insurance type, ZIP code	98%
Clinical History	Diagnoses (ICD-10), procedures, comorbidities	95%
Lab Results	Blood panels, vitals at admission/discharge	72%
Medications	Prescriptions at discharge, drug interactions	88%
Social Determinants	Income bracket, transportation access, caregiv	41%
Clinical Notes	Free-text physician notes (unstructured)	60%
Prior Visits	ER visits, outpatient visits in last 12 months	93%
Length of Stay	Days in hospital, ICU days	99%

Challenges to Pose *(Pick 2–3 during Structure phase)*

On missing data

"You mentioned [model]. How would you handle the fact that 41% of patients are missing social determinant data? Do you drop those features? Impute? Something else?"

On interpretability

"The CMO wants interpretability. You proposed [model]. How would you explain a specific patient's risk score to a cardiologist who has 5 minutes between rounds?"

On department differences

"Oncology has the highest readmission rate. But is that because oncology patients are sicker, or because the oncology department has a discharge problem? How would you disentangle that?"

On recall vs precision tradeoff

> *"You're optimizing for recall to catch at-risk patients. But if you flag too many, the intervention team gets overwhelmed and starts ignoring the alerts. How do you balance this?"*

The Twist *(Reveal around minute 18 of the case)*

DROP THIS AT 18:00

"One thing I should mention — the hospital recently started a pilot with a partner to digitize clinical notes using NLP. They've processed about 10,000 notes so far and found that physician sentiment in discharge notes (phrases like 'patient is not confident in self-care' or 'limited family support') is actually one of the strongest predictors of readmission. But it's only available for a small subset. How does this change your approach?"

What a strong answer looks like: Two-model approach or staged rollout. Base model uses structured data (available for all patients). NLP features boost predictions where notes exist. As digitization scales, the NLP model improves. Ship what works now — don't wait for perfect data.

What a Strong Candidate Covers

1. **Reframes the problem:** *"How do we identify high-risk patients before discharge so MediCare Plus can target interventions and reduce readmissions from 18% to below 14%?"*
2. **Asks about data completeness** — this is the key challenge in this case
3. **Addresses class imbalance** — class weights, threshold tuning, or SMOTE
4. **Chooses an interpretable model** given the CMO requirement — gradient boosted trees with SHAP, or logistic regression
5. **Has a strategy for missing data** — not just "impute," but which method and why, or models that handle missingness natively
6. **Handles the NLP twist** with a pragmatic staged approach
7. **Quantifies impact:** 120K admissions × 18% = 21,600 readmissions. Reducing to 14% saves 4,800 × $8,500 = $41M, minus $9M intervention cost = **$32M net savings**
8. **Addresses implementation:** EMR integration, clinician training, feedback loop for model monitoring

Timing Guide

Phase	Target Time	Notes
Read scenario aloud	0:00 – 2:00	
His reframe + questions	2:00 – 7:00	Show exhibits when relevant
His structure	7:00 – 20:00	Challenge at 12:00 and 16:00
Drop the twist	18:00	Watch how he adapts
Ask for synthesis	22:00	*"Can you wrap up with a recommendation?"*
Synthesis	22:00 – 25:00	

Coding Problems

Problem 1 — Patient Readmission Summary *(7 minutes)*

You have two CSV files:

- `patients.csv` — Patient demographics: `patient_id, age, gender, insurance, zip_code, chronic_conditions`
- `admissions.csv` — Hospital admissions: `admission_id, patient_id, admission_date, department, length_of_stay, discharge_disposition, er_visits_prior_12m, readmitted_30d`

Task: Write a Python function that takes the file paths as input and returns a DataFrame with one row per **department**, containing:

```
1. department — Department name
2. total_admissions — Total number of admissions
3. readmission_rate — Percentage with 30-day readmission (1 decimal)
4. avg_length_of_stay — Average LOS in days (1 decimal)
5. avg_patient_age — Average patient age (1 decimal)
6. pct_medicare — % of admissions with Medicare insurance (1 decimal)

→ Sort by readmission_rate descending. Merge on patient_id.
```

Problem 2 — High-Risk Patient Flagging *(7 minutes)*

Using the same datasets. A patient is **"high-risk"** if they meet **any two** of:

- They have **3 or more** chronic conditions
- They have been admitted **more than once**
- They have had **2 or more** ER visits in the prior 12 months

Return a DataFrame with: `patient_id, age, gender, chronic_conditions, total_admissions, max_er_visits, risk_flags`. Sort by `risk_flags` desc, then `total_admissions` desc.

Bonus: What percentage of high-risk patients had at least one readmission?

Code Comprehension — Conversational *(no coding required)*

Show him this code and ask the questions below:

```python
def analyze_readmissions(df):
    df['admission_date'] = pd.to_datetime(df['admission_date'])
    df = df.sort_values(['patient_id', 'admission_date'])

    df['prev_discharge'] = df.groupby('patient_id')['admission_date'].shift(1)
    df['days_since_last'] = (df['admission_date'] - df['prev_discharge']).dt.days
```

```
    result = df.dropna(subset=['days_since_last'])
    result = result[result['days_since_last'] <= 30]

    dept_counts = result.groupby('department').size().reset_index(name='rapid_readmissions')
    total_counts = df.groupby('department').size().reset_index(name='total')

    merged = dept_counts.merge(total_counts, on='department')
    merged['rapid_rate'] = (merged['rapid_readmissions'] / merged['total'] * 100).round(1)
    return merged.sort_values('rapid_rate', ascending=False)
```

Ask these three questions verbally:

1. **"Walk me through what this code does, line by line."**
   - Expected: Calculates rapid readmission rate by department using shift() to find gaps between consecutive admissions ≤ 30 days.
2. **"Is there a potential issue with this approach?"**
   - Expected: The shift-based method can over-count — a patient admitted Jan 1, Jan 20, Feb 15 would have Feb 15 flagged (26 days from Jan 20) even though it's 45 days from the original admission.
3. **"How would you improve this code?"**
   - Expected: Add docstrings, validate input types, parameterize the 30-day window, more explicit dropna handling.

Interviewer Rules

→ **Screen sharing required** for coding. He can use internet for syntax docs — no AI tools.

→ **Never volunteer information.** If he doesn't ask, don't tell.

→ **Challenge his model choices.** "Why not a neural network?" "What if the client can't interpret that?"

→ **Drop the twist** at 18 minutes and watch how he adapts.

→ **Use the scoring rubric** — fill it out during or immediately after.

→ **Keep it conversational.** You're a friendly DS interviewer, not an examiner.

Thank you for doing this. Make it real — that's how he gets better. After the debrief, share the scoring rubric so he can track progress across mocks. 🏔️